

## JUUPAJOKI HYYTIALA, Finland

WMO: 027700

Lat: 61.846N

Lon: 24.287E

Elev: 153

StdP: 99.50

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |            |            |            | MCWS/PCWD to 99.6% DB |            | WSF        |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|------------|------------|------------|-----------------------|------------|------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |            | 1%         |            |                       |            |            |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB       | WS         | MCDB       | MCWS                  | PCWD       |            |
| (a)<br>1      | (b)<br>-25.1 | (c)<br>-21.9 | (d)<br>-27.4                  | (e)<br>0.3 | (f)<br>-24.9 | (g)<br>-24.1 | (h)<br>0.4 | (i)<br>-21.9 | (j)<br>N/A            | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A | (n)<br>N/A            | (o)<br>N/A | (p)<br>N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.0                   | 26.8            | 17.7 | 24.7 | 16.9 | 22.7 | 16.0 | 19.6                | 24.1 | 18.4 | 22.5 | 17.2 | 21.2 | N/A                  | N/A  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.1                            | 13.3 | 20.9 | 16.8 | 12.2 | 20.0 | 15.6 | 11.3 | 18.7 | 56.6          | 24.4 | 52.6 | 22.3 | 48.8 | 21.3 | 24.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |       |      |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|-------|------|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |       |      |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      |       |      |
| (a) N/A           | (b) N/A | (c) N/A | (e) -27.0                  | (f) 28.6 | (g) 3.8            | (h) 2.7 | (i) -29.8                                          | (j) 30.5 | (k) -32.0  | (l) 32.1 | (m) -34.2  | (n) 33.6 | (o) -37.0  | (p) 35.5 |       |      |
| (4) N/A           | (5) N/A | (6) N/A | DB                         | WB       | -27.2              | 20.4    | 3.8                                                | 2.1      | -29.9      | 21.9     | -32.1      | 23.2     | -34.3      | 24.3     | -37.0 | 25.9 |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 4.3    | -8.0  | -6.0  | -2.8  | 3.0   | 9.6   | 13.3  | 16.5  | 14.7  | 10.1  | 3.9   | 0.2   | -3.9 |
|                                                                                  | DBStd       | 9.35   | 6.71  | 6.45  | 4.84  | 3.57  | 4.39  | 3.57  | 3.34  | 2.97  | 3.30  | 3.56  | 4.37  | 6.11 |
|                                                                                  | HDD10.0     | 2638   | 558   | 447   | 397   | 210   | 62    | 8     | 1     | 2     | 38    | 191   | 293   | 432  |
|                                                                                  | HDD18.3     | 5159   | 817   | 680   | 655   | 458   | 272   | 154   | 78    | 119   | 246   | 447   | 543   | 691  |
|                                                                                  | CDD10.0     | 553    | 0     | 0     | 0     | 2     | 49    | 109   | 202   | 147   | 42    | 2     | 0     | 0    |
|                                                                                  | CDD18.3     | 33     | 0     | 0     | 0     | 0     | 1     | 5     | 22    | 6     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 339    | 0     | 0     | 0     | 0     | 31    | 65    | 194   | 49    | 0     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 60     | 0     | 0     | 0     | 0     | 2     | 9     | 41    | 8     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Precipitation                                                                    | PrecAvg     | 600    | 46    | 33    | 31    | 30    | 38    | 68    | 78    | 70    | 54    | 58    | 48    | 46   |
|                                                                                  | PrecMax     | 841    | 89    | 69    | 64    | 60    | 89    | 107   | 142   | 163   | 141   | 131   | 120   | 97   |
|                                                                                  | PrecMin     | 418    | 12    | 2     | 5     | 3     | 6     | 21    | 22    | 6     | 9     | 3     | 5     | 12   |
|                                                                                  | PrecStd     | 86     | 21    | 18    | 14    | 16    | 20    | 24    | 33    | 38    | 35    | 35    | 25    | 20   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 3.3   | 5.3   | 9.6   | 18.9  | 26.6  | 28.0  | 30.0  | 27.7  | 20.7  | 13.2  | 9.9   | 6.9  |
|                                                                                  |             | MCWB   | 2.7   | 2.8   | 4.2   | 10.1  | 14.9  | 17.4  | 18.8  | 19.3  | 15.7  | 10.9  | 9.0   | 6.3  |
|                                                                                  | 2%          | DB     | 2.0   | 3.1   | 6.4   | 15.1  | 23.7  | 25.2  | 27.8  | 24.4  | 18.3  | 11.5  | 8.1   | 4.4  |
|                                                                                  |             | MCWB   | 1.5   | 1.9   | 2.9   | 7.8   | 14.0  | 16.1  | 18.8  | 17.3  | 14.3  | 10.1  | 7.5   | 3.8  |
|                                                                                  | 5%          | DB     | 0.8   | 2.0   | 4.5   | 12.0  | 21.2  | 22.8  | 25.5  | 22.3  | 16.5  | 10.5  | 6.6   | 3.0  |
|                                                                                  |             | MCWB   | 0.5   | 1.4   | 2.0   | 6.3   | 13.1  | 15.2  | 18.0  | 16.3  | 13.4  | 9.4   | 6.3   | 2.5  |
|                                                                                  | 10%         | DB     | 0.1   | 1.0   | 3.0   | 9.7   | 18.5  | 20.6  | 23.3  | 20.4  | 15.1  | 9.3   | 5.4   | 2.1  |
|                                                                                  |             | MCWB   | -0.1  | 0.5   | 1.2   | 5.2   | 11.3  | 14.3  | 17.3  | 15.5  | 13.0  | 8.3   | 5.0   | 1.7  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 3.1   | 3.4   | 5.2   | 10.7  | 16.7  | 19.5  | 21.5  | 20.4  | 16.2  | 11.7  | 9.1   | 6.6  |
|                                                                                  |             | MCDB   | 3.3   | 4.6   | 7.8   | 16.6  | 23.0  | 24.2  | 26.0  | 25.4  | 19.2  | 12.8  | 9.6   | 6.7  |
|                                                                                  | 2%          | WB     | 1.6   | 2.2   | 3.6   | 8.8   | 15.3  | 17.6  | 20.2  | 18.6  | 15.2  | 10.8  | 7.6   | 3.9  |
|                                                                                  |             | MCDB   | 1.9   | 2.9   | 5.6   | 13.9  | 21.2  | 22.7  | 24.8  | 22.3  | 17.3  | 11.3  | 8.1   | 4.4  |
|                                                                                  | 5%          | WB     | 0.6   | 1.4   | 2.5   | 7.1   | 13.8  | 16.3  | 19.2  | 17.4  | 14.3  | 9.6   | 6.3   | 2.6  |
|                                                                                  |             | MCDB   | 0.8   | 1.9   | 4.0   | 11.2  | 19.1  | 20.9  | 23.5  | 20.5  | 15.9  | 10.2  | 6.6   | 2.9  |
|                                                                                  | 10%         | WB     | 0.0   | 0.6   | 1.5   | 5.7   | 12.3  | 15.0  | 18.0  | 16.4  | 13.3  | 8.3   | 5.1   | 1.7  |
|                                                                                  |             | MCDB   | 0.2   | 1.0   | 2.7   | 9.0   | 17.8  | 19.4  | 22.1  | 19.3  | 14.6  | 9.1   | 5.3   | 1.9  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 5.7   | 5.8   | 8.3   | 9.7   | 11.6  | 10.2  | 10.0  | 9.6   | 7.8   | 5.6   | 3.7   | 4.1  |
|                                                                                  |             | MCDBR  | 3.4   | 5.1   | 9.1   | 14.8  | 16.3  | 14.2  | 13.2  | 12.7  | 10.1  | 7.0   | 4.4   | 4.0  |
|                                                                                  |             | MCWBR  | 3.4   | 4.4   | 6.2   | 8.5   | 7.8   | 6.5   | 5.6   | 6.4   | 6.3   | 5.5   | 4.2   | 3.9  |
|                                                                                  | 5% WB       | MCDBR  | 3.4   | 4.5   | 7.4   | 12.8  | 14.1  | 11.8  | 11.3  | 10.7  | 8.5   | 6.1   | 4.5   | 4.0  |
|                                                                                  |             | MCWBR  | 3.4   | 4.0   | 5.4   | 7.9   | 7.2   | 5.9   | 5.5   | 5.8   | 5.9   | 5.5   | 4.3   | 3.9  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.243  | 0.290 | 0.317 | 0.346 | 0.344 | 0.342 | 0.365 | 0.351 | 0.328 | 0.317 | 0.266 | 0.356 |      |
|                                                                                  | taud        | 2.064  | 2.153 | 2.232 | 2.394 | 2.469 | 2.497 | 2.450 | 2.480 | 2.538 | 2.411 | 2.165 | 2.609 |      |
|                                                                                  | Ebn at Noon | 441    | 662   | 789   | 846   | 878   | 885   | 851   | 834   | 792   | 646   | 414   | 286   |      |
|                                                                                  | Edh at Noon | 37     | 68    | 91    | 96    | 96    | 95    | 98    | 87    | 69    | 54    | 33    | 22    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.21   | 0.77  | 2.25  | 3.67  | 4.94  | 5.24  | 5.02  | 3.89  | 2.27  | 0.94  | 0.25  | 0.11  |      |
|                                                                                  | RadStd      | 0.04   | 0.18  | 0.27  | 0.41  | 0.55  | 0.41  | 0.51  | 0.44  | 0.29  | 0.18  | 0.05  | 0.01  |      |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (54 neighbors) | +0.49 | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | -132        | N/A     | N/A     | N/A     |

Nomenclature: See separate page